

ORF 405: Regression and Time Series

Jianqing Fan — Frederick L. Moore'18 Professor of Finance

Problem Set #1

Fall 2026

Due Saturday, September 12, 2026.

Instructions. Submit one PDF file. Show your work for mathematical problems. For the empirical problem, include concise code, relevant figures or tables, and a short interpretation.

1. Let $(\mathbf{X}, Y)$ denote a generic observation and let $f^*(\mathbf{x}) = \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x})$.

- (a) Let $\hat{f}$ be an estimator constructed from an independent training sample, and fix a testing point $\mathbf{x}_0$. Assume $Y_0 = f^*(\mathbf{x}_0) + \varepsilon_0$, where $E\varepsilon_0 = 0$ and $\text{var}(\varepsilon_0) = \sigma^2(\mathbf{x}_0)$. Show the bias-variance decomposition of the mean-square prediction error:

$$\mathbb{E}\{Y_0 - \hat{f}(\mathbf{x}_0)\}^2 = \sigma^2(\mathbf{x}_0) + \{\mathbb{E}\hat{f}(\mathbf{x}_0) - f^*(\mathbf{x}_0)\}^2 + \text{var}\{\hat{f}(\mathbf{x}_0)\}.$$

- (b) Explain briefly why flexible regression methods may have small bias but large variance.
(c) Define the population best linear predictor by

$$\boldsymbol{\beta}^* = \operatorname{argmin}_{\boldsymbol{\beta} \in \mathbb{R}^p} E\{Y - \mathbf{X}^T \boldsymbol{\beta}\}^2.$$

Assuming $\boldsymbol{\Sigma} = \mathbb{E}(\mathbf{X}\mathbf{X}^T)$ is nonsingular, show that $\boldsymbol{\beta}^* = \boldsymbol{\Sigma}^{-1}\mathbb{E}(\mathbf{X}Y)$ and that $\boldsymbol{\beta}^* = \operatorname{argmin}_{\boldsymbol{\beta} \in \mathbb{R}^p} E\{f^*(\mathbf{X}) - \mathbf{X}^T \boldsymbol{\beta}\}^2$. Thus, linear regression is the best linear approximation in mean squared prediction error and in approximating the true regression function f^* .

- (d) Let $(\mathbf{X}_i, Y_i)$, $i = 1, \dots, n$ be i.i.d. observations from the general regression problem. Derive directly, without using matrix notation, the least square estimator that minimizes $n^{-1} \sum_{i=1}^n (Y_i - \mathbf{X}_i^T \boldsymbol{\beta})^2$ is given by

$$\hat{\boldsymbol{\beta}} = \left(n^{-1} \sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i^T \right)^{-1} n^{-1} \sum_{i=1}^n \mathbf{X}_i Y_i.$$

- (e) By the law of large numbers, it holds under the mild conditions

$$n^{-1} \sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i^T \xrightarrow{P} \boldsymbol{\Sigma}, \quad n^{-1} \sum_{i=1}^n \mathbf{X}_i Y_i \xrightarrow{P} \mathbb{E}\mathbf{X}Y,$$

where $\xrightarrow{P}$ means convergence in probability as $n \rightarrow \infty$. Use this to show that the bias of the linear predictor $\hat{f}(\mathbf{x}_0) = \hat{\boldsymbol{\beta}}^T \mathbf{x}_0$ is approximately $f^*(\mathbf{x}_0) - (\boldsymbol{\beta}^*)^T \mathbf{x}_0$.

2. Consider the linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad E\boldsymbol{\varepsilon} = \mathbf{0}, \quad \text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

where $\mathbf{X}$ is an $n \times p$ full-column-rank matrix.

- (a) Show that the residual vector $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to every column of $\mathbf{X}$, namely the residual is uncorrelated with each covariate.

- (b) Prove the Gauss-Markov Theorem.
- (c) If $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, show that $\mathbb{E}\hat{\sigma}^2 = \sigma^2$ and $\text{var}(\hat{\sigma}^2) = 2\sigma^4/(n-p)$.
- (d) Consider the ANOVA problem. Let $\mathbf{P}_i = \mathbf{X}_i(\mathbf{X}_i^T \mathbf{X}_i)^{-1} \mathbf{X}_i^T$ be the projection matrices ($i = 1, 2$) with $\mathbf{X}_2$ has more columns than $\mathbf{X}_1$ (smaller model). Shows that the RSS reduction

$$\text{RSS}_0 - \text{RSS}_1 = \mathbf{Y}^T (\mathbf{P}_2 - \mathbf{P}_1) \mathbf{Y} \stackrel{\text{under } H_0}{=} \boldsymbol{\varepsilon}^T (\mathbf{P}_2 - \mathbf{P}_1) \boldsymbol{\varepsilon}$$

where $H_0 : \mathbf{Y} = \mathbf{X}_1 \boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and that $(\mathbf{I}_n - \mathbf{P}_2)(\mathbf{P}_2 - \mathbf{P}_1) = \mathbf{0}$.

- (e) If $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, show that $\text{RSS}_0 - \text{RSS}_1$ and RSS_1 are independent.

3. For observations (x_i, y_i) , $i = 1, \dots, n$, consider the simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

- (a) Show that

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

- (b) Show that $\text{var}(\hat{\beta}_1 | x_1, \dots, x_n) = \sigma^2 / \sum_{i=1}^n (x_i - \bar{x})^2$ if $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$.
- (c) Derive the above results in part (a) and (b) using the formulas given in the class with the matrix notation.
- (d) Define $\text{TSS} = \sum_i (y_i - \bar{y})^2$, $\text{RSS} = \sum_i (y_i - \hat{y}_i)^2$, and $\text{ESS} = \sum_i (\hat{y}_i - \bar{y})^2$. Prove that $\text{TSS} = \text{ESS} + \text{RSS}$. This is indeed a special case of problem 2(d).

4. **(Polynomial regression and cross-validation).** Generate data from

$$X_i \sim \text{Uniform}[-2, 2], \quad Y_i = 1 + 2X_i - X_i^2 + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2),$$

with $n = 200$.

- (a) Fit polynomial regressions of degrees 1, 2, $\dots$, 8 with $\sigma = 1$.
- (b) Use 5-fold cross-validation to choose the polynomial degree.
- (c) Repeat the experiment for two noise levels: $\sigma = 0.5$ and $\sigma = 2$.
- (d) Report the cross-validation errors in a table and plot the fitted curve for the selected model.
- (e) Discuss how the selected degree changes with the noise level.

5. **(Prediction of Housing Prices)** Zillow is an online real estate database company that was founded in 2006. An important task for Zillow is to predict the house price. However, their accuracy has been criticized a lot. According to Fortune, "Zillow has Zestimated the value of 57 percent of U.S. housing stock, but only 65 percent of that could be considered 'accurate' – by its definition, within 10 percent of the actual selling price. And even that accuracy isn't equally distributed". Therefore, Zillow needs your help to build a housing pricing model to improve their accuracy. Download the training and testing data:

<https://fan.princeton.edu/fan/classes/525/DataSets/train.data.csv>

<https://fan.princeton.edu/fan/classes/525/DataSets/test.data.csv>

and read the data (training data: 15129 cases, testing data: 6484 cases)

```

train.data <- read.csv('train.data.csv', header=TRUE)
test.data <- read.csv('test.data.csv', header=TRUE)
train.data$zipcode <- as.factor(train.data$zipcode)
test.data$zipcode <- as.factor(test.data$zipcode)

```

where the last two lines make sure that zip code is treated as factor. Letting $\mathcal{T}$ as a test set, define out-of-sample R^2 as of a prediction method $\{\hat{y}_i^{method}\}$ as

$$R^2 = 1 - \frac{\sum_{i \in \mathcal{T}} (y_i - \hat{y}_i^{method})^2}{\sum_{i \in \mathcal{T}} (y_i - \bar{y}^{train})^2},$$

where $\bar{y}^{train} = \text{ave}(\{y_i\}_{i \in \mathcal{T}_0})$ and $\mathcal{T}_0$ is the training set.

- (a) Calculate out-of-sample R^2 using variables “bathrooms”, “bedrooms”, “sqft_living”, and “sqft_lot”.
- (b) Calculate out-of-sample R^2 using the 4 variables above along with interaction terms.
- (c) Add the factor zipcode to (b) and compute out-of-sample R^2 .
- (d) Add the following additional variables to (d): $X_{12} = I(view == 0)$, $X_{13} = L^2$, $X_{13+i} = (L - \tau_i)_+^2$, $i = 1, \dots, 9$, where τ_i is $10 * i^{th}$ percentile and L is the size of living area (“sqft_living”). Compute out-of-sample R^2 .
- (e) Now without assessing to the test data, using the 5-fold cross-validation and the training data alone, compute the out-of-sample R^2 for the last model.